

Assignment 8

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Complexity Analysis React Program

To filter students enrolled in a specific course, the following operation is performed in the program:

```
const filteredStudents = filterCourse
  ? sortedStudents.filter((student) =>
    student.enrolledCourses.has(filterCourse)
  )
  : sortedStudents;
```

Let:

- **n** = total number of students stored in the system.
- Each student contains a **Set of enrolled courses**.

The filtering process works as follows:

1. The filter() function iterates through all students in the sortedStudents array.
2. For each student, the program checks whether the given course exists in the student's enrolledCourses set using:

```
student.enrolledCourses.has(filterCourse)
```

3. Since enrolledCourses is implemented as a **Set**, the has() operation has an average time complexity of **O(1)** (constant time).

Therefore, for **each of the n students**, the program performs a constant-time membership check.

Thus, the total time complexity becomes:

$$T(n) = n \times O(1)$$
$$T(n) = O(n)$$

Final Result

The **time complexity of filtering students by course** is:

$$O(n)$$

where **n is the number of students in the system.**

This is efficient because the algorithm scans the list of students only once and uses constant-time lookup in a Set to check course membership.